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M1 Macro
Group A
Session 14

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

FOC $\quad \beta b \tilde{L}_{t+1} - (\alpha_1 + b + \beta b)\tilde{L}_t + b\tilde{L}_{t-1} = 0$

$\hookrightarrow \quad \tilde{L}_t = c_1 \lambda_1^t + c_2 \lambda_2^t$

- $0 < \lambda_1 < 1, \quad \lambda_2 > 1$

$$\lambda_1 = \frac{(\alpha_1 + b + \beta b) - \sqrt{(\alpha_1 + b + \beta b)^2 - 4\beta b^2}}{2\beta b}$$

$\hookrightarrow p(\lambda) = \beta b \lambda^2 - (\alpha_1 + b + \beta b)\lambda + b$

$$\lambda_2 = \frac{\cdots + \sqrt{\cdot}}{\cdots}$$

c_1? c_2? $\longrightarrow$ boundaries condition

$=$ $\hat{I}_0$ and $\hat{I}_\infty$ are given

- $\hat{I}_0 = c_1 \lambda_1^0 + c_2 \lambda_2^0 = c_1 + c_2$ 1 eq with c_1 and c_2

- $\lim_{t \to \infty} \hat{I}_t = 0$ $\longrightarrow$ I converge back to the SS.

$\Rightarrow \lim_{t \to \infty} \underbrace{c_1 \lambda_1^t}_{\to 0} + c_2 \underbrace{\lambda_2^t}_{\to \infty} = 0$ 1 eq with c_1 & c_2

$\Longrightarrow c_2 = 0$

Therefore $c_2 = 0$

$$c_1 + c_2 = \hat{L}_0 \Rightarrow c_1 = \hat{L}_0$$

The optimal solution is :

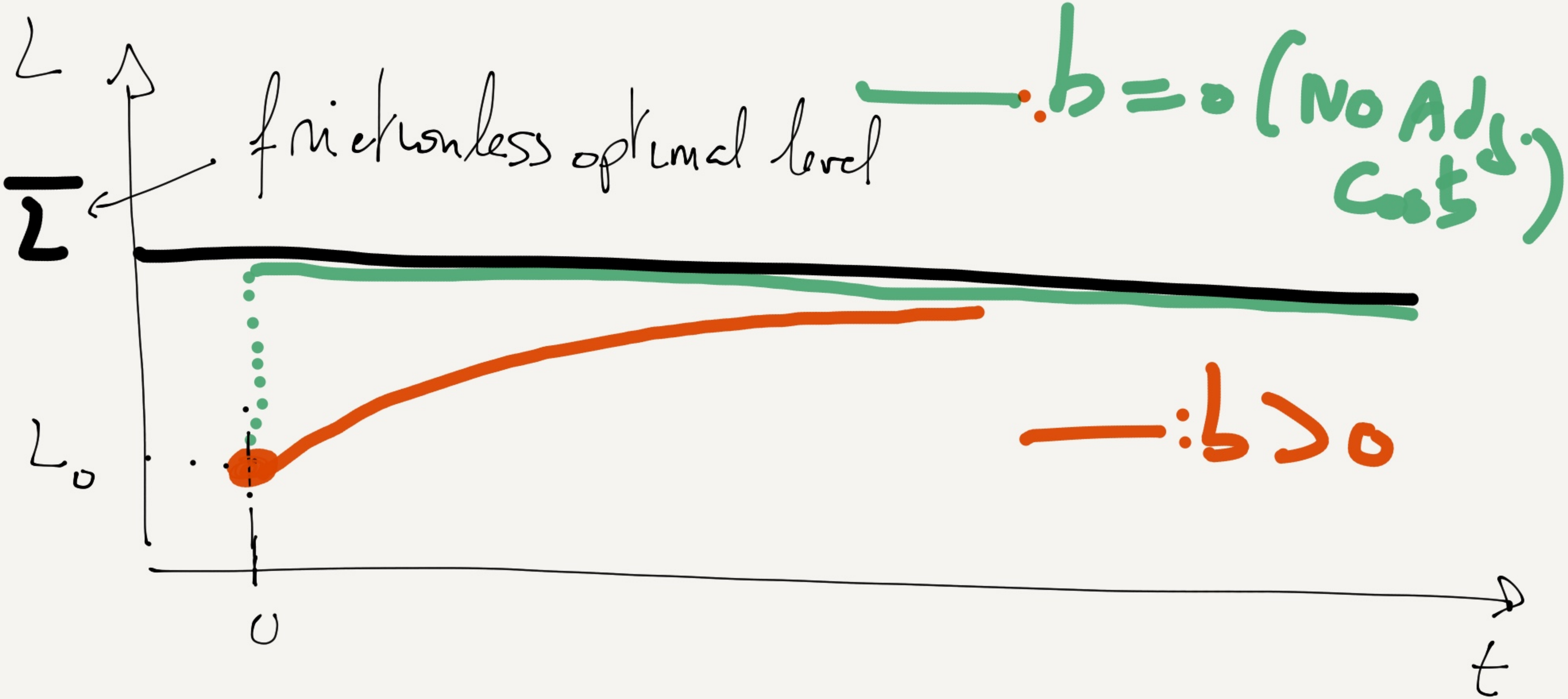
for $t \geq 1$: $\hat{L}_t = \hat{L}_0 \times \lambda_1^t$

$$= \lambda_1 \times \hat{L}_0 \lambda_1^{t-1}$$

$$\boxed{\hat{L}_t = \lambda_1 \hat{L}_{t-1}}$$

$$\Rightarrow L_t - \bar{L} = \lambda_1 (L_{t-1} - \bar{L}) \Rightarrow \boxed{L_t = (1-\lambda_1)\bar{L} + \lambda_1 L_{t-1}}$$

$$\boxed{L_t = (1-\lambda_1)\bar{L} + \lambda_1 L_{t-1}}$$



If $b = 0$ (no adj costs)

FOC $\quad \cancel{\beta b \bar{L}_{t+1}} - (\alpha_1 + \cancel{b + \beta b})\bar{L}_t + \cancel{b \bar{L}_{t-1}} = 0$

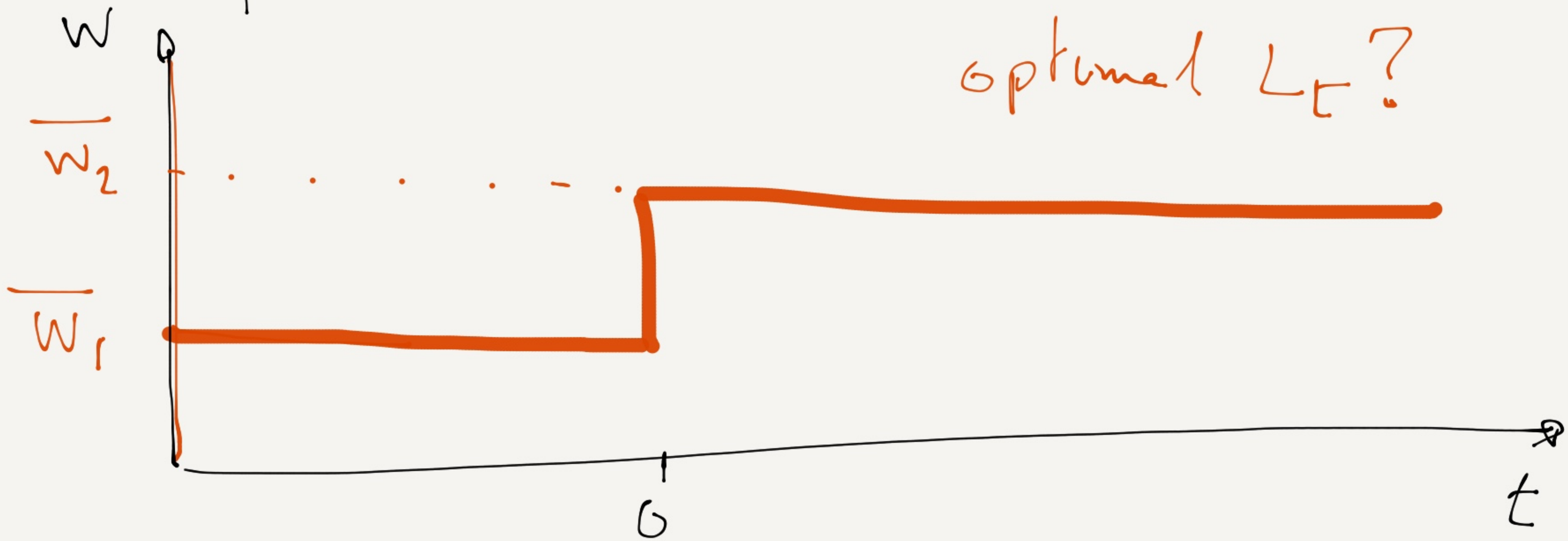
$$\Rightarrow \quad \alpha_1 \bar{L}_t = 0$$

$$\alpha_1 (L_t - \bar{L}) = 0$$

$$\boxed{L_t = \bar{L}}$$

An example of optimal path

- for $t<0$ $\overline{w} = \overline{w}_1$
- in $t=0$, unexpectedly, $w_0 = \overline{w}_2$
  for all $t \geq 0$, $w_t = \overline{w}_2 > \overline{w}_1$



optimal L_t is $L_t = (1-\lambda_1)\overline{L} + \lambda_1 L_{t-1}$

for $t<0$, $\overline{L} = \overline{L_1} = \frac{\alpha_0 - \overline{W_1}}{\alpha_1}$

$L_t = \overline{L_1}$ (because it has been for ever in the past)

$L_{-1} = \overline{L_1}$

for $t \geq 0$ $\overline{L} = \overline{L_2} = \frac{\alpha_0 - \overline{W_2}}{\alpha_1} < \overline{L_1}$

and $\boxed{L_t = (1-\lambda_1)\overline{L_2} + \lambda_1 L_{t-1}}$

$$L_{-1} = \overline{L}_1 = \frac{\alpha_0 - \overline{W}_1}{\alpha_1}$$

$$L_0 = (1-\lambda_1)\,\overline{L}_2 + \lambda_1 L_{-1}$$

$$= \boxed{(1-\lambda_1)\,\overline{L}_2 + \lambda_1 \overline{L}_1}$$

$$L_1 = (1-\lambda_1)\,\overline{L}_2 + \lambda_1 L_0 = (1-\lambda_1)\,\overline{L}_2 + \lambda_1 (1-\lambda_1)\,\overline{L}_2 + \lambda_1^2 \overline{L}_1$$

$$L_1 = (1-\lambda_1)(1+\lambda_1)\,\overline{L}_2 + \lambda_1^2 \overline{L}_1$$

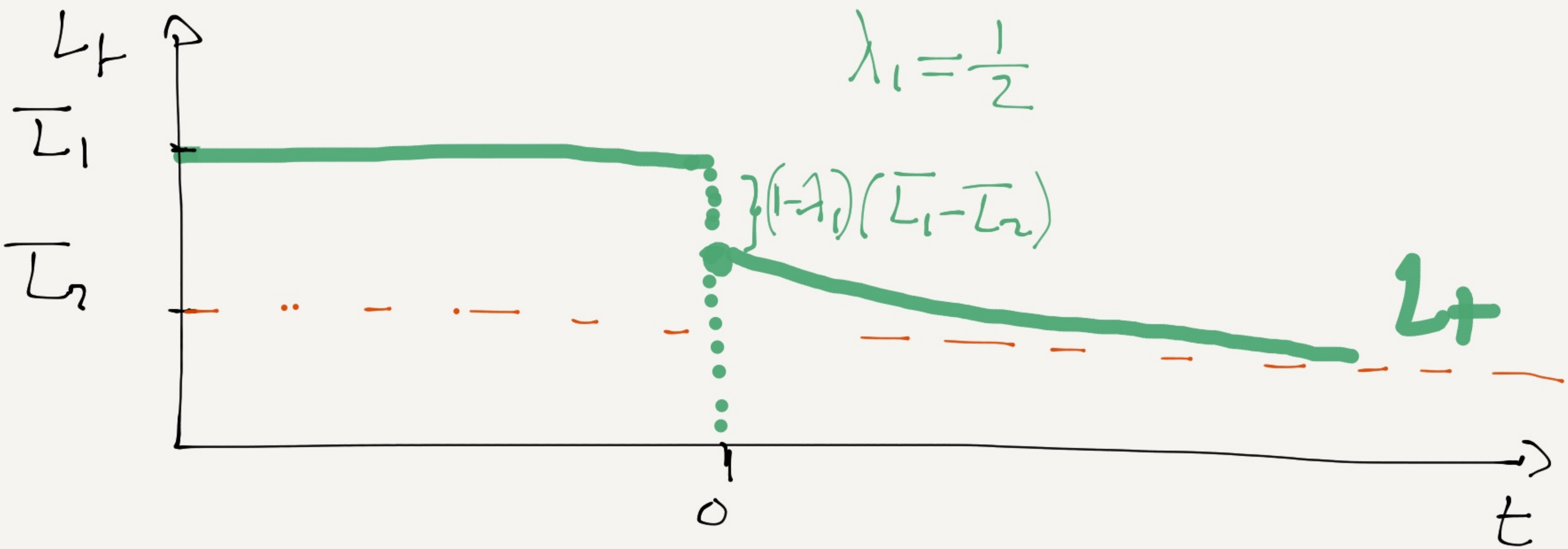
$$L_2 = (1-\lambda)\,\overline{L}_2 + \lambda_1 L_1 = (1-\lambda_1)\,\overline{L}_2 + (1-\lambda_1)(1+\lambda_1)\lambda_1 \overline{L}_2 + \lambda_1^3 \overline{L}_1$$

$$= (1-\lambda_1)(1+\lambda_1+\lambda_1^2)\,\overline{L}_2 + \lambda_1^3 \overline{L}_1$$

$$\cdots \quad L_t = (1-\lambda_1)(1+\lambda_1+\lambda_1^2+\cdots+\lambda_1^{t-1})\,\overline{L}_2 + \lambda_1^t \overline{L}_1$$

$$\lim_{t\to\infty} L_t = (1-\lambda_1)\underbrace{\left(\sum_{i=0}^{\infty} \lambda_1^i\right)}_{\frac{1}{1-\lambda_1}} \bar{L}_2 + \underbrace{\left[\lim_{t\to\infty} \lambda_1^t\right]}_{0} \times \bar{L}_1$$

$$\Rightarrow \lim_{t\to\infty} L_t = \bar{L}_2$$



The role of future wages

Another way of solving the FOC

$$\hat{L}_{t+1} = \frac{\alpha_1+b+\beta b}{\beta b}\hat{L}_t - \frac{1}{\beta}\hat{L}_{t-1} + \hat{W}_t \quad (1)$$

add the equation $\hat{L}_t = 1 \times \hat{L}_t + 0 \times \hat{L}_{t-1}$ (2)

- (1) and (2) under a matrix form :

$$\underbrace{\begin{pmatrix} \hat{L}_{t+1} \\ \hat{L}_t \end{pmatrix}}_{X_{t+1}} = \underbrace{\begin{bmatrix} \frac{\alpha_1+b+\beta b}{\beta b} & -\frac{1}{\beta} \\ 1 & 0 \end{bmatrix}}_{A} \underbrace{\begin{pmatrix} \hat{L}_t \\ \hat{L}_{t-1} \end{pmatrix}}_{X_t} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \hat{W}_t$$

The dynamics is driven by the eigenvalues of A, which are the solutions of

$$\det(A - \lambda I) = 0$$

$$A - \lambda I = \begin{pmatrix} \frac{\alpha_1 + b - \beta b}{\beta b} - \lambda & -\frac{1}{\beta} \\ 1 & -\lambda \end{pmatrix}$$

$$\det(A - \lambda I) = 0 \iff \lambda^2 - \frac{\alpha_1 + b - \beta b}{\beta b}\lambda + \frac{1}{\beta} = 0$$

Characteristic polynomial we already studied

- A has 2 eigenvalues λ_1, λ_2 with

$0 < \lambda_1 < 1$

$\lambda_2 > 1$